Public Economics (ECON 131)
Section #7: Capital Income and Savings Taxation (EXTRA)

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Contents

1 Savings Taxation (Cont.)

1.1 Practice Problems

1.1.1 Guber, Ch. 22, Q.13 (Modified)

Consider the model covered in Section 7 again in which an individual lives for two periods and has the following utility function: $U(C_1, C_2) = \ln C_1 + \ln C_2$. However, instead of setting parameters, let us solve for a more general case. In other words, suppose an individual earns some income W in the first period and saves with an interest rate r to finance consumption in the second period.

(a) Set up the individual's lifetime utility maximization problem. Solve for the optimal C_1 and C_2.

first, what's the IBC

IBC: $W = C_1 + S = C_1 + \frac{C_2}{1+r}$

where $C_2 = S(1+r)$

$\Rightarrow S = \frac{C_2}{1+r}$

via substitution $\Rightarrow W = C_1 + \frac{C_2}{(1+r)}$

$\Rightarrow W - C_1 = \frac{C_2}{1+r}$

$\Rightarrow (W - C_1)(1+r) = C_2$

maximize

$U = \ln(C_1) + \ln(C_2)$

$U = \ln(C_1) + \underbrace{\ln([W - C_1][1+r])}_{\ln(W - C_1) + \ln(1+r)}$

$\frac{\partial U}{\partial C_1} = \frac{1}{C_1} + \frac{1}{W - C_1}(-1) = 0$

$\Rightarrow \frac{1}{W - C_1} = \frac{1}{C_1}$ $\Rightarrow C_1 = W - C_1$

$\Rightarrow 2C_1 = W$

$\Rightarrow C_1^* = \frac{W}{2}$

$C_2^* = \left(W - \frac{W}{2}\right)(1+r) = \frac{W}{2}(1+r)$

(b) Suppose the government imposes a 20% tax on interest income. Solve for the new optimal levels of C_1 and C_2.

What changes? C_2 is now $C_2 = (W - C_1)[1 + (1 - .2)r] = (W - C_1)(1 + 0.8r)$

IBC is now $W = C_1 + \frac{C_2}{1+0.8r}$

$$\max U = \ln(C_1) + \underbrace{\ln[(W-C_1)(1+0.8r)]}_{\ln(W-C_1) + \ln(1+0.8r)}$$

$$\frac{\partial U}{\partial C_1} = \frac{1}{C_1} + \frac{1}{W-C_1}(-1) = 0$$

$$\Rightarrow \frac{1}{C_1} = \frac{1}{W-C_1}$$

$$\Rightarrow W - C_1 = C_1$$

$$\Rightarrow W = 2C_1$$

$$\Rightarrow C_1^* = \frac{W}{2}$$

same as above!

$$C_2 = (W-C_1)(1+0.8r)$$

$$C_2 = \left[W - \frac{W}{2}\right](1+0.8r)$$

$$C_2^* = \frac{W}{2}(1+0.8r)$$

1.2 Adding labor decisions (Model with 3 variables)

Consider again the model above in which an individual lives for two periods, but now we will add labor supply to the individual's decision. So, let's assume that this individual now has the following utility function: $U(C_1, C_2, l) = \ln C_1 + \ln C_2 + \ln(L - l)$. Suppose the individual earns a wage w for each unit of labor supplied in period one, denoted as l. They again can save for their retirement with an interest rate r.

(a) Set up the individual's lifetime utility maximization problem. Solve for the optimal C_1, C_2, and l.

what is the IBC?

income = consumption

$$w \cdot l = C_1 + S$$

$$w \cdot l = C_1 + \frac{C_2}{1+r}$$

$$\Rightarrow (w \cdot l - C_1)(1+r) = C_2$$

Lagrangian

$$\mathcal{L} = \ln(C_1) + \underbrace{\ln[(wl - C_1)(1+r)]}_{\ln(wl - C_1) + \ln(1+r)} + \ln(L - l)$$

$$\frac{\partial \mathcal{L}}{\partial C_1} = \frac{1}{C_1} + \frac{1}{wl - C_1}(-1) = 0 \Rightarrow \frac{1}{C_1} = \frac{1}{wl - C_1}$$

$$\Rightarrow wl - C_1 = C_1$$

$$\Rightarrow wl = 2C_1$$

$$\Rightarrow \frac{wl}{2} = C_1$$

$$\frac{\partial \mathcal{L}}{\partial l} = \frac{1}{wl - C_1} \cdot w + \frac{1}{L - l}(-1) = 0$$

$$\Rightarrow \frac{1}{L - l} = \frac{w}{wl - C_1}$$

substitute in

$$\Rightarrow wl - C_1 = w(L - l)$$

$$\Rightarrow wl - \frac{wl}{2} = wL - wl$$

$$\Rightarrow \frac{wl}{2} + wl = wL$$

$$\Rightarrow \frac{3}{2}\cancel{w}l = \cancel{w}L$$

$$\Rightarrow l^* = \frac{2}{3}L$$

from above: $C_1 = \frac{w}{2} l$

$$\Rightarrow C_1 = \frac{w}{\cancel{2}} \cdot \frac{\cancel{2}}{3} L$$

$$\Rightarrow C_1^* = \frac{1}{3} wL$$

Last step: plug l^* & C_1^* to find C_2^*

$$C_2 = \left[w \cdot \frac{2}{3}L - \frac{1}{3}wL\right](1+r)$$

$$C_2^* = \frac{1}{3}wL(1+r)$$